Karan Kapur

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Education

University of Wisconsin-Madison Madison, WI
Bachelor of Science in Computer Science and Data Science, Minor in Entrepreneurship *May 2026*

Experience

Data Analyst 08/2026 - Present
Wisconsin Union *Madison, WI*

- **Replaced the team's old scheduling software with an app I shipped in 2 days, used daily by 250+ employees**: 4,400+ shifts, self-serve accounts, and work chat; built on TypeScript, MySQL, Docker, and AWS.
- **Forecasted weekly food demand and staffing** from sales and shift data in Excel to guide orders and schedules.

Co-Founder & Cloud Storage Engineer 04/2026 - Present
thaw *Madison, WI*

- **Cut LLM session cold boots on H100s ∼386x, from ∼340s to a 0.88s median fork**: co-founded an open-source inference startup (Rust, CUDA) with N. Matteson and M. Yu.

VP, National Executive Council 03/2026 - Present
Kappa Eta Kappa *Remote*

- **Converted a 50+ school TAM into 2 chartered and installed chapters**: built a **Python** scraping pipeline that found universities without a professional engineering fraternity, sourced contacts, and ran the **outbound**.

Software Engineer 01/2026 - 05/2026
Eurofins Scientific *Madison, WI*

- **Shipped a forecasting product lab technicians used daily**: partnered with Eurofins as a client through UW-Madison, interviewed technicians and chemists across **4+ business units**, and took it to **production**.
- **Turned 10K+ messy records into forecasts with confidence scores**: built the ETL pipelines, **PostgreSQL** schema, **Python FastAPI** backend, and **React** frontend; iterated on operator feedback.

Automation & Compliance Engineering Intern 07/2024 - 09/2024
HCLSoftware *Noida, India*

- **Lifted onboarding conversion 30%** via an analytics-driven redesign; 50+ tests shortened regression 60%.

Projects

Delivery Doctor (solo build) | Agent telemetry transport verifier | github.com/karank2512/delivery-doctor 10/2026

- **Reproduced 9 delivery failure modes end to end**: canary replay read back from intake catches auth, expired-key, schema, buffer, network, rate-limit, too-large, timeout, and overflow failures.

Foreman (solo build) | **Agentic** workforce platform | github.com/karank2512/Foreman 09/2026

- **Lowered cost per run 30% vs an all-LLM pipeline on a 100-record paired benchmark**: routed 5 of 7 steps to deterministic code, taking LLM calls per run from 11 to 6 (list prices).
- **Ran every agent in propose mode**: workers research and draft, nothing leaves the workspace without a human approver, and slipping workers get replaced by versioned candidates.

RelayIQ (solo build) | **Agentic** decision engine | github.com/karank2512/RelayIQ 07/2026

- **Held writes to exactly once under retries and duplicate deliveries**: idempotent **webhook** handling and an auditable SQL ledger of every credit spent.
- **Achieved 3.3x lower cost per usable lead on a 495-case seeded benchmark**: built a multi-step agent that routes each field to the best-fit **enrichment** provider API and only spends a credit when it's worth it.

Technical Skills

Languages and Stack: **Python**, **TypeScript**, **SQL**, Rust, C++, Next.js, React, FastAPI, PostgreSQL, Docker, AWS
AI and Agents: LLM tool calling, human approval, LLM-judge **evals**, RAG, MCP, Claude Code and Codex daily
GTM and Tools: **HubSpot** APIs, **webhooks**, exactly-once processing, GitHub Actions **CI**, Clay (familiar)